

NANTUCKET, MA, USA

WMO: 725060

Lat: 41.253N

Lon: 70.061W

Elev: 48

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 14756

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	18.2	-1.9	5.0	16.7	2.0	6.1	20.2	36.6	33.3	33.2	33.4	15.5	310	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	79.5	70.7	77.2	70.2	75.1	69.2	73.5	76.8	72.3	75.2	71.2	73.7	11.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	120.3	75.4	71.7	117.1	74.3	70.3	111.7	72.9	37.1	77.0	36.0	75.1	35.1	73.9	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.8	26.7	24.3	DB	8.5	85.6	5.4	3.0	4.6	87.7	1.5	89.5	-1.5	91.1	-5.4	93.3	(4)
			WB	6.5	76.0	5.0	1.5	2.9	77.2	0.0	78.1	-2.8	78.9	-6.5	80.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	33.2	33.6	37.8	45.8	54.0	62.0	69.1	69.2	64.1	55.4	46.6	38.6
	DBStd	14.14	8.60	6.92	6.41	4.85	4.89	4.82	3.79	3.83	4.82	6.13	6.83	7.65
	HDD50	2041	523	458	379	142	17	0	0	0	21	142	359	
	HDD65	5498	987	878	842	577	342	113	9	8	73	300	551	818
	CDD50	2364	0	0	1	15	141	360	592	596	424	189	40	6
	CDD65	349	0	0	0	0	1	23	136	140	46	3	0	0
	CDH74	724	0	0	0	3	5	52	314	305	42	3	0	0
	CDH80	66	0	0	0	0	1	7	35	21	2	0	0	0
Wind	WSAvg	12.3	14.0	13.9	14.3	13.2	11.9	10.8	9.8	9.3	10.7	12.7	13.2	13.5
Precipitation	PrecAvg	40.0	3.8	3.3	3.2	3.3	3.4	2.5	2.4	2.8	3.5	3.3	3.7	4.6
	PrecMax	57.2	7.4	5.6	6.3	7.2	10.4	6.9	6.6	6.7	9.5	7.4	7.4	9.7
	PrecMin	25.3	1.2	0.7	0.7	1.3	0.6	0.2	0.1	0.3	0.1	0.9	1.2	1.7
	PrecStd	9.7	1.4	1.4	1.6	1.5	2.5	1.8	1.9	1.7	2.3	1.7	1.6	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	51.6	55.6	63.4	72.5	79.6	83.8	82.7	77.5	71.9	63.4	58.5
		MCWB	51.4	49.5	49.9	52.6	62.2	68.1	72.9	72.0	69.5	65.9	59.5	55.6
	2%	DB	49.9	48.2	51.8	57.5	67.3	74.7	80.2	79.3	74.9	68.4	61.1	55.1
		MCWB	48.3	45.8	47.5	50.8	59.2	66.3	70.9	71.6	68.7	64.4	58.2	53.3
	5%	DB	47.7	46.0	49.0	55.1	63.9	71.9	77.3	77.2	73.1	66.1	58.9	52.9
		MCWB	45.6	43.9	45.8	49.8	57.3	65.0	70.1	71.0	68.1	63.1	55.9	51.1
	10%	DB	45.4	44.2	46.9	53.3	61.5	69.5	75.0	75.2	71.7	64.2	56.9	50.3
		MCWB	43.4	41.6	44.1	48.8	56.0	64.2	69.1	69.9	67.1	61.2	54.1	48.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.6	49.9	52.2	56.7	64.4	71.3	75.3	76.1	73.0	68.0	61.5	56.5
		MCDB	51.9	50.8	53.7	59.9	70.1	77.2	80.5	79.8	75.6	69.9	62.3	57.6
	2%	WB	48.7	46.6	48.8	53.7	61.8	68.5	73.3	74.1	71.2	65.7	58.9	53.8
		MCDB	49.3	47.4	50.3	55.5	65.7	72.3	77.3	77.0	73.3	67.1	60.5	54.6
	5%	WB	46.2	44.7	46.5	51.6	59.3	66.8	71.9	72.8	69.9	64.1	56.8	51.4
		MCDB	47.1	45.8	48.0	53.4	62.5	69.9	75.2	75.3	71.7	65.4	58.5	52.4
	10%	WB	43.7	42.1	44.7	49.9	57.3	65.2	70.8	71.8	68.6	62.3	54.6	48.4
		MCDB	45.2	43.9	46.5	52.0	60.1	67.8	73.5	74.0	70.4	64.0	56.3	50.0
Mean Daily Temperature Range	5% DB	MDBR	11.4	11.6	11.3	11.8	12.0	11.4	11.3	11.5	12.0	11.5	11.4	11.7
		MCDBR	12.5	13.4	13.0	14.6	15.8	15.3	13.9	12.4	12.4	11.9	12.0	12.8
		MCWBR	12.9	12.5	10.7	9.5	9.4	8.9	7.3	7.1	8.3	9.5	11.2	13.7
	5% WB	MCDBR	12.5	12.9	12.0	12.3	14.0	12.4	11.4	10.4	10.5	10.6	11.1	13.1
		MCWBR	13.4	13.1	11.3	10.0	9.5	8.2	7.1	6.9	8.3	10.0	11.5	14.5
Clear-Sky Solar Irradiance	taub	0.321	0.327	0.345	0.384	0.418	0.465	0.487	0.447	0.377	0.357	0.328	0.316	
	taud	2.494	2.481	2.464	2.368	2.313	2.220	2.167	2.281	2.443	2.501	2.544	2.539	
	Ebn at Noon	265	281	287	282	273	259	252	259	272	266	260	257	
	Edh at Noon	24	28	32	37	40	44	46	40	32	27	23	21	
All-Sky Solar Radiation	RadAvg	534	828	1178	1534	1737	1870	1913	1728	1411	962	624	458	
	RadStd	44	78	96	163	170	165	103	102	108	79	51	44	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+187	+92

Nomenclature: See separate page